Case 21: Coupang App Usage Dropped 15%

Interview Prompt — App usage has dropped 15%. Investigate the drop and recommend actions.

Step 1: Clarify

I'd start with a few clarifying questions, because "app usage" is not a single metric:

- **What is being measured?** DAU, MAU, sessions per user, time spent per session, or app opens? Each points to a different problem.
- **What is the comparison?** MoM, YoY, or against a specific pre-period? Korea has strong seasonality and a MoM read after a major sale event will always look negative.
- **What is the shape of the drop?** A step change on a specific date, or a gradual decline over weeks?
- **Is it platform-wide or concentrated** by OS, app version, geography, or customer cohort?
- **Did anything change** — an app release, a tracking or SDK update, an OS-level privacy change, a membership pricing change, a competitor launch?
- **What happened to orders and GMV** in the same period? Usage without a revenue read is not yet a business problem.

Assume: DAU declined 15% MoM, platform-wide, with orders down ~6% over the same period.

Step 2: Build the Metric Tree

DAU = Installed base × Activation rate (share of installs opening on a given day)

And usage intensity decomposes as:

Total sessions = DAU × Sessions per active user

For a retention-style read, DAU is better decomposed additively:

DAU(t) = New users + Retained users + Resurrected users − Churned users

Component	Previous	Current	Change
New	0.6M	0.5M	−0.1M
Retained	8.2M	6.9M	−1.3M
Resurrected	1.2M	1.1M	−0.1M
DAU	**10.0M**	**8.5M**	**−1.5M (−15%)**

Retained users account for ~87% of the decline. This is not an acquisition problem — existing users are opening the app less.

Step 3: Establish the Shape of the Decline

Before segmenting, I'd plot daily DAU across the full period, because the shape narrows the hypothesis class:

- **Step change on a single date** → app release, outage, tracking change, or a policy change
- **Gradual weekly decay** → cohort quality, competitive pressure, or product experience degradation
- **Weekday/weekend divergence** → use-case change rather than a technical fault

Suppose: the decline is a step change beginning on the 8th of the month, holding flat at the lower level afterwards.

That rules out gradual causes and points at something that happened on or just before that date.

Step 4: Rule Out Measurement Before Behaviour

A step change on a specific date is as likely to be a tracking artefact as a real behavioural shift. I'd check first:

- Was an app version released on or near that date? What is the analytics SDK version in it?
- Does the drop appear in server-side order and session logs, or only in the client-side analytics?
- Did the definition of a "session" or an "active user" change in the pipeline?
- Any OS-level change (an iOS or Android privacy update) affecting attribution or background events?

Suppose: server-side session logs confirm the decline is real, and orders are down ~6%, so this is behaviour, not instrumentation.

Step 5: Segment the Problem

Contribution analysis across dimensions, isolating only those contributing disproportionately to the −1.5M:

Dimension	Cuts
Platform	iOS vs Android, by app version
Membership	WOW member vs non-member
Cohort	Tenure (new / 1–6M / 6M+), frequency tier (heavy / medium / light)
Geography	Metro vs non-metro
Entry point	Push notification, organic open, deep link, search

Dimension	Cuts
Surface	Rocket Delivery, Eats, Play, Fresh

Suppose the analysis shows:

Segment	DAU Change
Android	−4%
iOS	−27%
WOW members	−8%
Non-members	−19%
Push-originated opens	−41%
Organic opens	−6%

Two signals now stand out: iOS specifically, and push-originated sessions specifically.

Step 6: Funnel and Entry-Point Analysis

For a usage metric the relevant funnel is the *entry* funnel, not the purchase funnel:

Push sent → Push delivered → Push opened → Session started → Browse → Order

Stage	Before	After
Push sent	40M	40M
Push delivered	38M (95%)	22M (55%)
Push opened	4.2M	2.4M
Session from push	4.0M	2.3M

Push *delivery* has collapsed, while send volume is unchanged. The problem is upstream of user behaviour entirely.

Step 7: Validate Root Cause

I'd partner with Engineering and CRM to check:

- Push service provider logs and error rates by OS
- The app release notes for the version shipped on the 8th
- Notification permission opt-in rates by cohort
- Whether iOS notification categories or entitlements changed in the release

Suppose we find: the release on the 8th changed the notification payload structure, causing a large share of iOS pushes to fail delivery silently. Non-members were disproportionately affected because they rely more on promotional pushes to re-enter the app, while WOW members open organically out of habit.

Root cause identified. The 15% DAU decline is primarily a push-delivery failure on iOS following a specific app release, not a demand or competitive problem.

Step 8: Quantify Before Recommending

Reconciling the pieces:

- Lost push-originated sessions: ~1.7M/day
- Of the −1.5M DAU, push-originated loss explains the large majority; residual ~0.2M is organic softness
- Orders down ~6% against usage down 15% — implying the lost sessions were disproportionately low-intent browsing, so revenue impact is real but sub-proportional

This matters for the recommendation: the fix is urgent, and the business impact is smaller than the headline metric suggests.

Recommendation

Immediate

- Roll back or hotfix the notification payload change; validate delivery rates on iOS before release
- Backfill the affected window with a corrected re-engagement campaign to the impacted cohort, weighted toward non-members
- Add push delivery rate by OS to the daily operating dashboard

Medium-Term

- Automated anomaly detection on push delivery rate, session starts by entry point, and DAU by OS — alerting on threshold breach rather than waiting for a monthly review
- Release-health checks that gate deployment on delivery and session metrics, not only on crash-free rate
- Reduce structural dependence on push for non-member re-entry by improving organic re-engagement surfaces

Long-Term

- Canary releases with metric-based rollback triggers
- Segment-level early warning: WOW member DAU is the health signal for habit, non-member DAU is the signal for marketing-driven traffic — track them separately, because a blended number masks exactly this kind of problem

Manager-Level Summary

> App usage fell 15% MoM, driven almost entirely by retained users rather than acquisition. The decline was a step change beginning on the 8th, which traced to an app release that broke notification payload delivery on iOS — push delivery fell from 95% to 55%, removing roughly 1.7M push-originated sessions a day. Non-members were hit hardest because they depend on push to re-enter the app, while WOW members continued opening organically. Orders declined ~6%, so the revenue impact is real but sub-proportional to the usage drop, since the lost

sessions skewed low-intent. I'd hotfix the release immediately, run a corrected re-engagement campaign to the affected cohort, and add push delivery rate and entry-point session mix to release-health gating so this class of failure is caught in hours rather than at month end.